

08) k-Nearest Neighbours

⑧

① Normalised data:

Customer	Support tickets	usage (hours/month)	cancelled
A	0.2	0.8	0
B	0.9	0.15	1
new	0.6	0.4	

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$\begin{aligned} A &= \sqrt{(0.6 - 0.2)^2 + (0.4 - 0.8)^2} \\ &= \sqrt{0.4^2 + (-0.4)^2} = \sqrt{0.32} = 0.565 \end{aligned}$$

$$\begin{aligned} B &= \sqrt{(0.6 - 0.9)^2 + (0.4 - 0.15)^2} \\ &= \sqrt{(-0.3)^2 + (0.25)^2} = \sqrt{0.09 + 0.0625} \\ &= \sqrt{0.1525} = 0.39 \end{aligned}$$

new customer distance from A: 0.565

new customer distance from B: 0.39

c) The new customer is closer to B so the prediction will be: yes, he will cancel

12) I can break the tie by changing the k to odd number (5,3) or by weighting the votes by distance so closer neighbours count more.

$$\sqrt{(8.0 - 1.0)^2 + (5.0 - 1.0)^2} = 7.0$$

$$\sqrt{(8.0 - 1.0)^2 + (5.0 - 2.0)^2} = 7.29$$

$$7.0 < 7.29 \Rightarrow (1.0, 1.0) = \text{yes}$$

$$\sqrt{(7.0 - 1.0)^2 + (6.0 - 2.0)^2} = 6.32$$

$$\sqrt{(6.0 - 1.0)^2 + (5.0 - 2.0)^2} = 5.10$$

$$6.32 < 5.10 \Rightarrow (1.0, 2.0) = \text{yes}$$

The new customer is closer to A than B, so the prediction will be: yes, he will cancel